

REVIEW OF THE CRITICAL VELOCITY OF GAS-PLASMA INTERACTION

I: *Experimental Observations*

LARS DANIELSSON

Royal Institute of Technology, Stockholm, Sweden

(Received 30 April, 1973)

Abstract. A review is given of the experimental investigations concerning the critical velocity of the interaction between a neutral gas and a plasma in relative motion. In most of the experiments this critical velocity is equivalent to a voltage limitation of a discharge through a partially ionized, magnetized plasma. The critical velocity phenomenon can have been of importance in a large number of experiments but it has been observed in rather few cases and studied in detail in less than half a dozen plasma machines. The major investigations were made in rotating plasma devices like the Homopolar, plasma guns and a plasma – neutral gas impact experiment. The emphasis of this paper is concentrated, though not limited, to collision-free plasmas. Thus the (MPD)-arc experiments are not extensively treated. It is concluded that the existence, under certain conditions, of a critical velocity, critical voltage or critical E/B (depending on the particular observation) is proved by sufficient experimental evidence.

In a following article in this issue by J. Sherman the theoretical work in the same field is discussed.

1. Introduction

The critical velocity, or – to connect more directly to the actual observations – the voltage limitation of an electric discharge in the presence of a magnetic field, is a well known observational fact which has been reported many times and from several different types of discharges. The oldest reports seem to be by Simon in 1959 on a PIG discharge and independently by Alfvén and Fahleson in 1960–1961 on a homopolar-like (rotating) plasma device. Most observations have been accidental and the effect has appeared as an unwanted limitation of the energy storage in various plasma devices, for example thermonuclear machines like the Ixion, the early Homopolars and Lehnert's F-machines. Special arrangements have also been made to avoid this unwanted limitation and these have to some extent been successful. In most cases this has been accomplished by, as far as possible, keeping the plasma and the neutral gases apart. As far as we have found only four experiments have attempted to study the voltage limitation effect in itself, viz. the ones by Angerth *et al.* (1962), Eninger (1965), Lehnert *et al.* (1966), and Danielsson (1970). These will be discussed in more detail below.

The critical velocity (voltage limitation) phenomenon is observed in electrical discharges where the discharge voltage and current are applied across a magnetic field. It appears as an upper limit of the burning voltage and this voltage level is directly proportional to the magnetic field strength but independent of gas and plasma pressures as well as discharge current. However, a necessary condition for the effect is that

there is neutral gas present; once a state of complete ionization is accomplished this limitation is no longer applicable. It can be pointed out already here that the ionization has to be complete *everywhere* in the volume of magnetically confined plasma before this happens. Another necessary condition is that the plasma and the gas are in relative motion with respect to one another.

The concept of the critical velocity comes directly from the above mentioned direct relationship between the observed voltage and magnetic field

$$(V/d) B = \text{const} = v_c$$

or, more correctly,

$$V = V_0 + v_c dB, \quad (1)$$

where d is a distance characteristic for the discharge and the constant has the dimension of velocity. The voltage, V , also contains a contribution, V_0 , due to electrode effects. On the local scale the velocity constant must also be directly related to the actual electric field in the plasma by

$$E/B = v_c \quad \text{or} \quad \mathbf{E} \times \mathbf{B}/B^2 = \mathbf{v}_c. \quad (2)$$

By means of probe measurements this has been observed in a few of the experiments to be discussed. The value of the parameter v_c is reported to depend only on the particular gas used in the experiment and it has been found that v_c can be estimated from the relation

$$\frac{1}{2} M v_c^2 = e V_i, \quad (3)$$

where M is the mass of the ion produced in the discharge, v_c is the critical relative velocity between the plasma and the gas and eV_i is the ionization energy of the gas. The formula (3) for the value of the critical velocity has been suggested hypothetically by Alfvén (1954). The value of v_c calculated from (3) for a few interesting gases is listed in Table I. (A few cases where the observed voltage level was reported not to be compatible with (3) will be discussed below.) As indicated in Table I the value of v_c is not unambiguous for molecular gases. Even more complicated is the situation for gas mixtures, but this case will not be covered in the present review.

Among the various types of discharges where the critical velocity or voltage limitation has been reported in the literature we can mention rotating plasma devices, plasma guns and shock tubes, and finally an experiment exhibiting a direct collision between a plasma and a neutral gas, which is a more direct application of the critical velocity concept. Table II summarizes the experiments where a critical velocity or a voltage limitation has been observed and discussed. One striking aspect of these experiments, directly evident from this table, is the wide range of configurations, gases, pressures and currents; the pressures over which the effect is observed vary from 1 μHg to 200 mmHg and the currents from below 1 A to 5×10^4 A, e.i. in both cases by five orders of magnitude.

TABLE I
Values of the critical velocity.

Atom or molecule	Created ion	Ionized mass	Ionization potential (V)	Critical velocity (km s ⁻¹)
H	H ⁺	1	13.6	51.0
H ₂	H ₂ ⁺	2	15.4	38.6
H ₂	H ⁺	1	18.0	59.0
He		4	24.5	34.4
C		12	11.3	13.5
N	N ⁺	14	14.6	14.2
N ₂	N ⁺	14	24.5	18.4
N ₂	N ₂ ⁺	28	15.6	10.4
O		16	13.6	12.8
Ne		20.2	21.6	14.4
Ar		39.9	15.6	8.7
Fe		55.8	7.8	5.2

2. Rotating Plasma Experiments

The voltage limitation effect has been observed in quite a few rotating plasma experiments; first, in a PIG discharge by Simon (1959), the Homopolar III by Bratenahl *et al.* (1959, 1960) and a homopolar-like device by Alfvén (1960) and Fahleson (1961). Later Angerth *et al.* (1962) made a detailed study of the phenomenon in this latter machine. From a Homopolar-like device the effect has also been reported by Barber *et al.* (1963). From the Ixion and Ixion-like devices, observations of the voltage limitation have been reported by Baker and Hammel (1961), Bannenberg *et al.* (1963) and Shvets *et al.* (1963). Finally, in Lehnert's F-machine the critical velocity still plays an important role and its experimental aspects have been extensively discussed by Bergström *et al.* (1962, 1963, 1966), Lehnert (1966), and Lehnert *et al.* (1966). As previously mentioned the voltage limitation can be a severe restriction for thermonuclear experiments. However, in many of these experiments the main body of the plasma is fully ionized, which means that it is possible to choose a magnetic field and vessel geometry so that the interaction between the plasma and the neutral gas (e.g. in the vicinity of the walls) is avoided, minimized or at least delayed. These efforts have been partly successful – the voltage limitation restriction removed – see for example Halbach *et al.* (1962) in the Homopolar V (Puffatron) or Shvets *et al.* (1966) in an Ixion-like device.

2.1. SIMON'S PIG DISCHARGE

Simon (1959) discussed the current-voltage characteristic of a PIG discharge. The arrangement is shown in Figure 1. The magnetic field was 5–15 kG with a mirror ratio of 2. A voltage of 30 kV was applied between anode and cathode. The resulting current and voltage obtained with hydrogen gas at 1 μ Hg pressure and magnetic field of 10 kG is shown in Figure 2. It is seen that the voltage first rises to a high value (about

TABLE II
Summary of the experiments

Author	Year	Type of experiment	Diam. $\times$ length cm	Gas	Pressure μ Hg	Magn. field kG	Current A	Discharge duration ms	Burning voltage kV	Plasma velocity km s ⁻¹
Simon	1959	PIG	15 $\times$ 70	H ₂	1	2-18	1-100	1	0.5-2.5	60
Alfvén	1960	Rotating		H ₂						
Fahlson	1961	plasma	40 $\times$ 15	He Ne A	5-200	1-10	30-10 ⁴	0.1	1-7	10-60
Angerth	1962			D ₂ N ₂ O ₂						
Bratenahl	1960	Homopolar	100 $\times$ 60	H ₂ D ₂	20-500	2.5-5	10 ³ -10 ⁵	0.5	2-3	
Baker	1961	Ixon	25 $\times$ 30	D ₂	2-5	4-10	10 ³ -10 ⁵	0.3	1-7	130
Bergström	1962	Rotating	50 $\times$ 40	H ₂ N ₂	5-50	1-4	10-10 ⁴	1	0.5-2	150
Lehnert	1966	plasma								
Bannenber	1963	Rotating	35 $\times$ 150	H ₂	5	4	low-10 ⁴	1	2	30
		plasma								
Barber	1963	Rotating	13.4 $\times$ 2.5	N ₂	40-4 $\times$ 10 ³	18	low	0.1	1	14
		plasma								
Shvets	1963	Rotating	12.5 $\times$ 180	H ₂	10 $\times$ 100	2	3000	0.3	0.5	60
		plasma								
Wilcox	1964	Coaxial	7.2 $\times$ 60	He A	5-2 $\times$ 10 ⁵	0.7-13	35-5500	0.1	0.1-1	10-60
Eniger	1965	gun		H ₂ N ₂						
Patrick and	1965	Coaxial	7.5 $\times$ 60	H ₂	20-200	3-5	5 $\times$ 10 ⁴	0.03	0.5	30-90
Pugh		shock								
Patrick and	1966	Coaxial	5 $\times$ 3	A H ₂ N ₂	200	0-10	400-1600	DC (10 ⁴)	0.05-0.1	
Schneiderman		arc		NH ₃						
Blackstock	1968	MHD-arc	4 $\times$ 4	Cs	$\geq$ 100	0.2-1	400-900	DC	0.015	2.4
Danielsson	1970	Plasma impact	5 $\times$ 5	He	3	2-5	0	2-20 $\times$ 10 ⁻³	0.3-8	30-500

20 kV) then quickly drops to a constant level of about 1800 V while the current decreases exponentially from about 100 A. The variation of this voltage with the magnetic field is reported to be $V \propto B^{4/5}$, i.e. almost linear. The first experimental point is at 2 kG and it is doubtful whether it is correct to extrapolate the voltage to zero for zero magnetic field instead of assuming a linear relation, cf. Figure 6 and 16. The value of the parameter $v = E/B = V/dB$ is for this case reported to be 60 km s^{-1} , i.e. near to v_c for hydrogen. It is also mentioned in Simon's report that the voltage level is independent of the gas pressure but no experimental data corroborating that observation are given in Simon's paper.

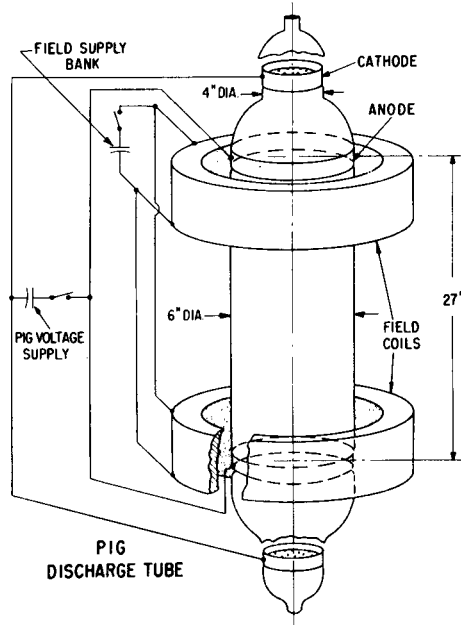


Fig. 1. Schematic drawing of PIG tube and electrical circuit. (Simon, 1959).

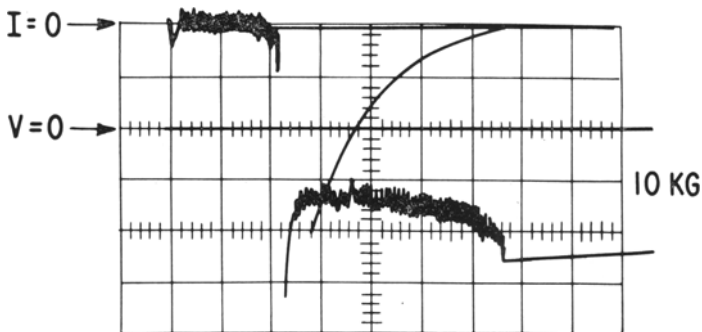


Fig. 2. Oscilloscope traces of the voltage and current in the discharge. The upper trace is the current which decays exponentially with time; the lower shows the voltage staying almost constant with time (Simon, 1959).

2.2. THE HOMOPOLARS

The design principles of a homopolar-like machine are shown in Figure 3. (from Angerth *et al.*, 1962). The electrodes are usually two coaxial cylinders, the magnetic field is axial with a small mirror ratio – between 1 and 2 – and the discharge current is radial. The Lorentz force will make the plasma rotate in the azimuthal direction. In the applications to be discussed here, the discharge vessel is pre-filled with the gas to be studied.

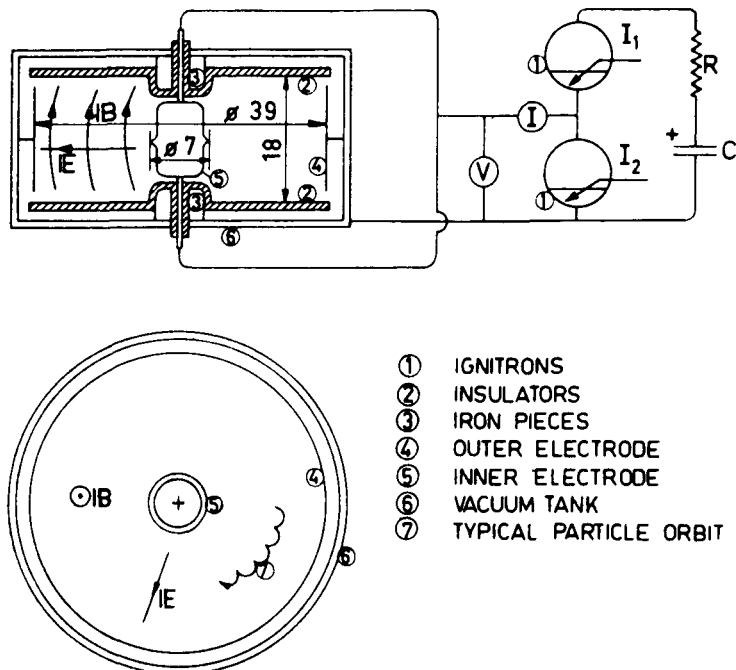


Fig. 3. Experimental arrangement of Angerth *et al.* (1962).

In the experiment described in three consecutive papers by Alfvén (1960), Fahleson (1961), and Angerth *et al.* (1962) the voltage limitation was first observed accidentally (as in all ‘thermonuclear’ experiments) but then investigated in detail. The main parameters of the experiment are displayed in Table II. The resistor R in Figure 3 was introduced specifically for this investigation to limit the current and the degree of ionization. By varying this resistor between 1 and 200 Ω , the bank voltage between 5 and 15 kV and the gas pressure between 5 and 200 μHg , the current-voltage characteristic was studied for a wide range of parameters in seven light gases. The observations were essentially similar to those of Simon (Figure 2) and the investigation gave the following results as far as the burning voltage level, V_b , was concerned:

(i) V_b was almost independent of pressure and current down to a minimum pressure and up to a maximum current. At the limit the degree of ionization approached 100%. See Figures 4 and 5.

(ii) V_b was independent of the power input.

(iii) V_b was directly proportional to magnetic field and the electrode separation d :

$$V_b \simeq v_c dB.$$

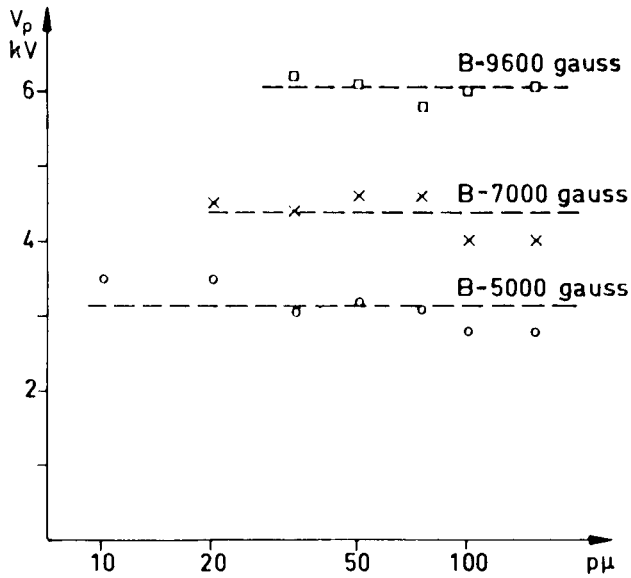


Fig. 4. Burning voltage as a function of pressure for hydrogen discharges (Angerth *et al.*, 1962)

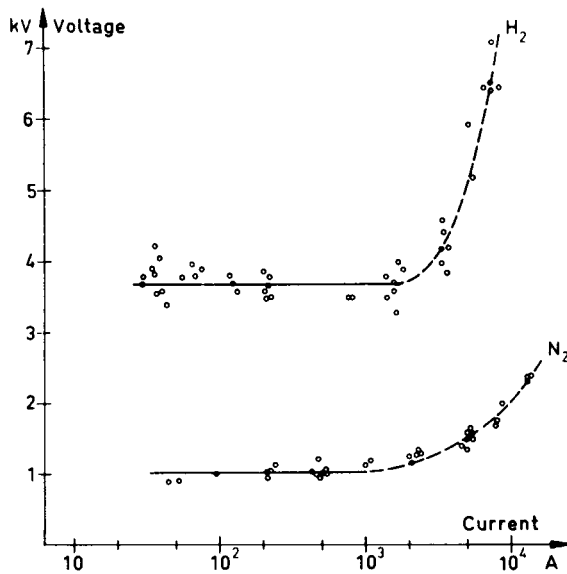


Fig. 5. Burning voltage versus current for hydrogen and nitrogen discharges. $B_0 = 6000$ G, $P = 30 \mu\text{Hg}$. (Angerth *et al.*, 1962)

Additional right-hand-side terms (electrode sheaths, centrifugal effect) were negligible in most cases. See Figure 6.

(iv) The proportionality constant v_c – the critical velocity – depended on, and only on, the gas used, and for all seven gases studied it was very close to the value according to (3).

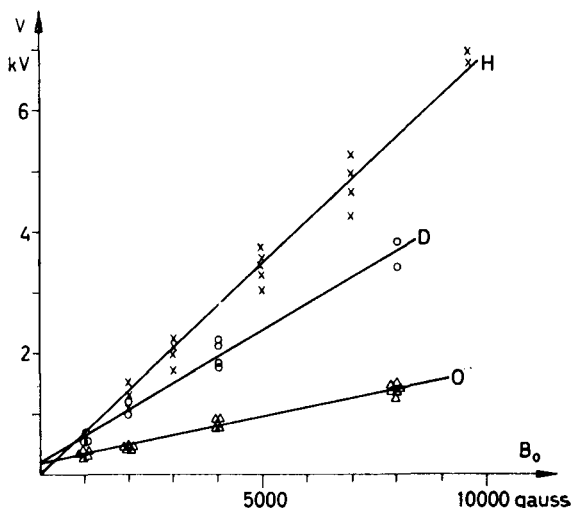


Fig. 6. Burning voltage vs magnetic field for H_2 , D_2 and O_2 (Angerth *et al.*, 1962).

In order to obtain a voltage-current relation like the one implied in Figure 2 for other gases than hydrogen it proved necessary in this experiment to bake the electrodes. Unless this was done the later part of the voltage plateau was elevated to the neighbourhood of that for hydrogen. This was interpreted as hydrogen outgasing from the electrodes. An attempt to verify the interpretation of the burning voltage level as the velocity of rotation was made by means of Doppler shift measurements. These were of a rather complicated character but they seemed to indicate that the impurity ions in a hydrogen discharge moved azimuthally with the velocity v_c for hydrogen everywhere while the neutral gas was almost stationary. However, as shown by Sherman in Part II of this review, the plasma ions could not have moved azimuthally with the velocity v_c ; due to the small ion Hall Parameter $(\omega\tau)_i$ the plasma ions moved essentially radially along the E -field under typical conditions.

The device used by the Stockholm group in the experiment described above was similar to the first three Berkeley Homopolars. A limitation of the energy storage was also observed in the Homopolars I, II and III. In the Homopolar III work reported by Bratenahl *et al.* (1959, 1960) this was related to the critical velocity and found to confirm very well the results of the Stockholm group. However, no detailed investigation of the phenomenon was made. Instead all efforts were directed towards avoiding the limitation. This was accomplished in the Puffatrons (Homopolars IV and V) by injecting the gas to be used through an electro-magnetic valve into the centre of the

discharge region and moving the material walls away from this region. Thus there was no background of neutral gas in the system and the plasma did not reach the insulators for a considerable time. The result – reported by Halbach *et al.* (1962) – was that no limitation of the rotational energy was observed during the initial stage of the discharge. Velocities up to 600 km s^{-1} were reported. Later, when plasma reached the wall, it was no longer possible to maintain this high velocity.

From investigations with a homopolar-like machine Barber *et al.* (1963) reported a current-voltage characteristic with a voltage plateau, which was increased with the magnetic field. This effect was, however, not given further attention and it is not clear to us whether this observation has any connection to the critical velocity – though this should be expected due to the essential similarity to the previous experiments and the actual level of the plateau seems to be about right.

2.3. THE IXIONS

Topologically the Ixions are similar to the usual PIG discharges. The main difference is that the discharge current is much larger in the Ixion. The design of the Ixion III of Los Alamos is shown in Figure 7. As in the PIG machines the discharge runs between a plasma column, acting as a centre cathode, and a coaxial anode. The main param-

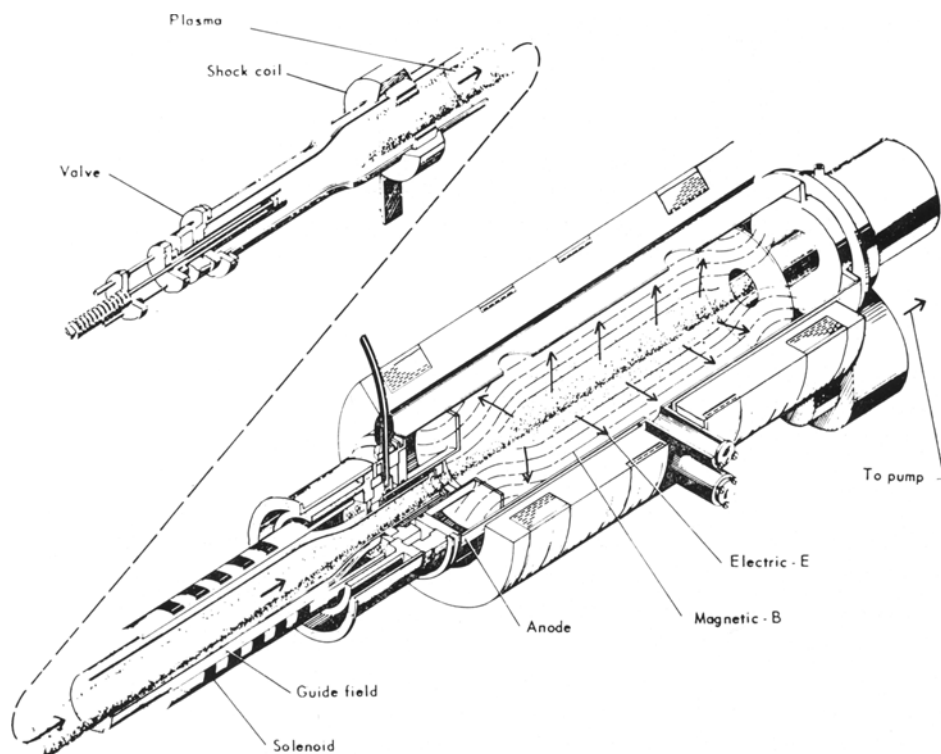


Fig. 7. Ixion III (Boyer *et al.*, 1958).

eters of the discharge are found in Table II. The mirror ratio is 2. In the mode of operation of interest to us in the present context the accelerating capacitor bank was disconnected after the acceleration phase and various loads or a high voltage boost bank applied subsequent to the disconnection. The application of this boost bank showed the existence of a limit of the rotating velocity and this was shown to be characteristic of the plasma and not controlled by, for example, voltage breakdown at the boundary insulators. When the boost voltage was increased the voltage across the discharge first increased but very soon reached a limiting value which was directly proportional to the magnetic field (see Figure 8). The current-voltage characteristic

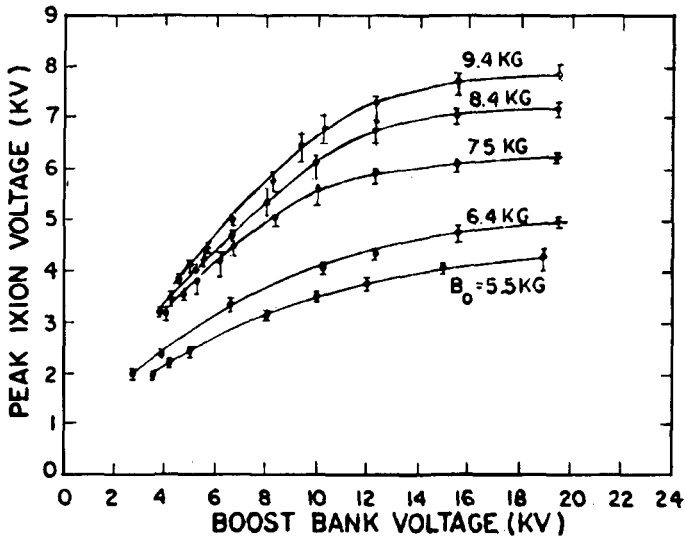


Fig. 8. Peak Ixion voltage attained under boost with magnetic field as a parameter (Baker *et al.*, 1961 II).

for the same operating conditions are shown in Figure 9. Thus it is obvious that the Ixion III displayed a strong voltage saturation and the ratio of the saturation voltage to the magnetic field was reported to be $8.3 \times 10^3 \text{ m}^2 \text{ s}^{-1}$ (0.83 V G^{-1}) or in terms of the critical velocity $v_c = E/B = 130 \text{ km s}^{-1}$ according to a measured and scaled voltage distribution in the midplane of the apparatus. Thus the limiting velocity in this case was considerably higher than predicted by (3) and this was true for such late times that the fully ionized plasma must long have been in contact with the end-insulators in the mirror region of the magnetic field. However, considering the data presented by Baker *et al.* (1961) and particularly their measured voltage distribution (see Figure 10) it seems clear that the E/B -value was not constant over the cross-section of the discharge. The value quoted referred to an inner section, while the value in the vicinity of the outer electrode was approximately $v_c = E/B = 45 \text{ km s}^{-1}$, i.e. subcritical if the distribution shown in Figure 10 is supposed to be applicable in the present case. At the end insulators, where the main interaction with neutral gases is likely to take place, the

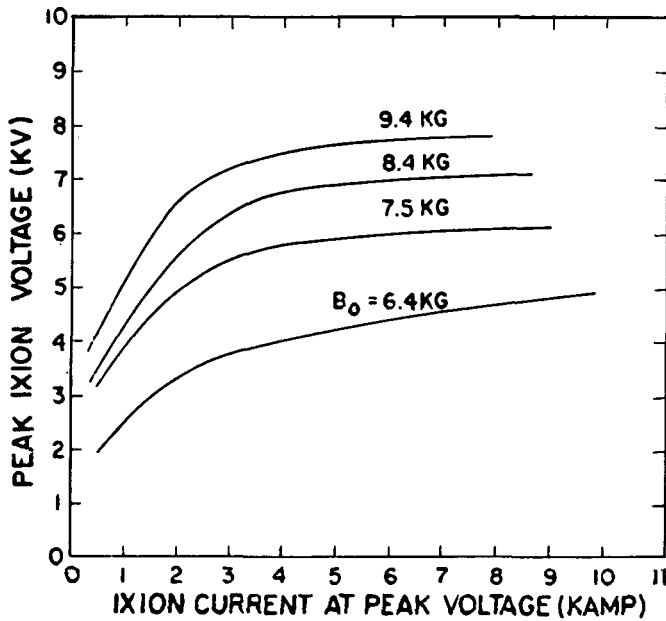


Fig. 9. Ion current-voltage characteristics with magnetic field as a parameter (Baker *et al.*, 1961 II).

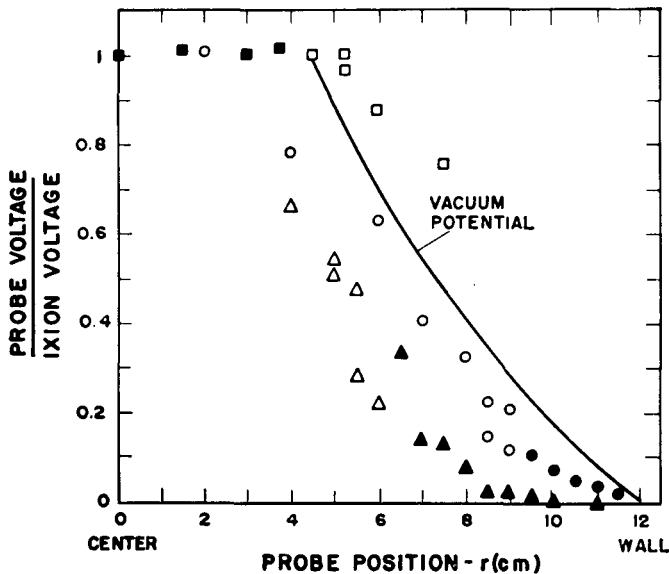


Fig. 10. Radial voltage distribution as obtained with an electrostatic probe. The circular and square data points give the voltage at the midplane. The triangular points give the voltage data at a station 19 cm from the midplane. Open points represent data taken at positions where serious probe perturbations were evident (Baker *et al.*, 1961 I).

local value of E/B was probably different from the value in the midplane (Figure 10) but no measurements were made here. If the velocity of rotation at the insulators is assumed to have been limited to the critical value (51 km s^{-1}) then isorotation of the plasma would allow an approximately 40% higher velocity in the midplane (mirror ratio = 2). Accounting for this and maybe only partial isorotation the value of E/B (or v_ϕ) was probably supercritical only in a central part of the discharge where there are many reasons to suppose that the plasma was fully ionized and hence no direct interaction with the neutral gases could take place. A similar situation has been observed and discussed by Lehnert *et al.* (1966), see below.

Bannenberg *et al.* (1963) and Shvets *et al.* (1963) have used devices similar to the Ixions. They both worked with hydrogen and burning voltage levels corresponding to about 30 km s^{-1} and 60 km s^{-1} respectively were observed. That is, in the first case the critical velocity for hydrogen was never reached. The limitation was, however, quite clear and one can speculate that it was determined by impurities. In both cases the voltage (velocity) limitation has later been avoided by removing the insulators from the vicinity of the discharge (Bannenberg *et al.*, 1966; Shvets *et al.*, 1966). Bannenberg was able to observe the velocity of rotation directly as a phase shift in the magnetic signals from probes which differed in azimuthal position only. He found the observed velocity to agree with the E/B -velocity.

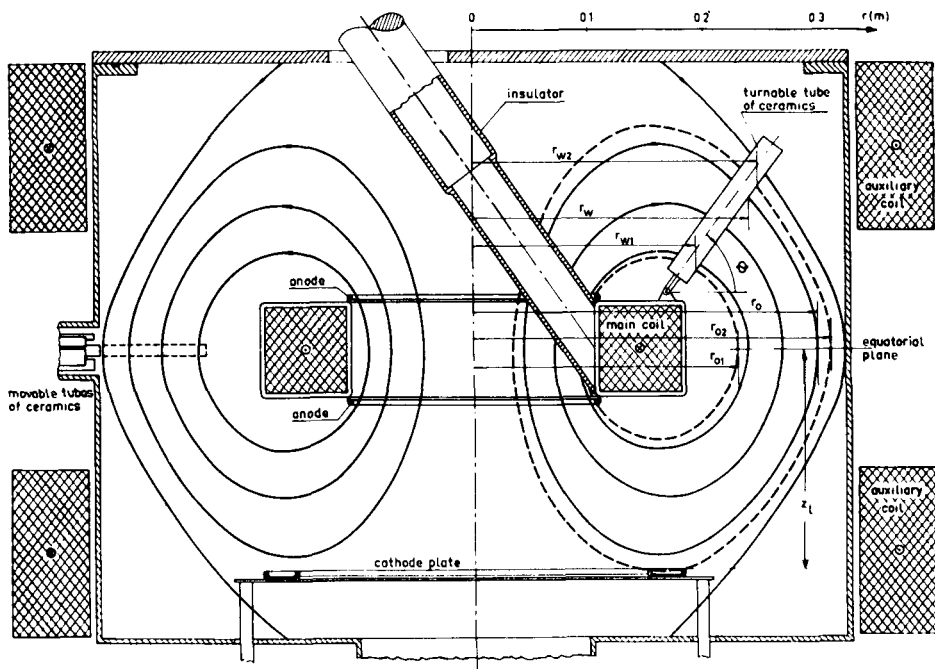


Fig. 11. The discharge chamber with the magnetic field configuration of device F II. The discharge takes place between two anode rings and a cathode plate. The confinement region is situated between the dashed lines in the right hand part of the figure (Lehnert *et al.*, 1966).

2.4. LEHNERT'S EXPERIMENTS

The voltage limitation effect has been of decisive importance in most of the F-machines of Lehnert *et al.* The main investigation of the phenomenon was done on the FII-device (Lehnert *et al.*, 1966) which is shown in Figure 11. The parameters of the discharge are found in Table II. The discharge takes place between the two anode rings and the cathode plate, across the magnetic field which has a maximum magnetic field strength of about 0.6 Vs m^{-2} in the equatorial plane. The mirror ratio is about 4. The location of the electrodes limited the plasma to within the dotted boundary in Figure 11. The Lorentz force made this fully ionized plasma rotate around the axis of the system. Like in the Ixion experiment we are here interested in a situation where the plasma was post-accelerated by a second condenser bank.

The sequence of oscillograms in Figure 12 demonstrates what happened when the energy of the second bank was increased. The voltage across the discharge, Φ_{12} , first increases with the bank energy but only up to certain limiting value, Φ_{12c} . After this

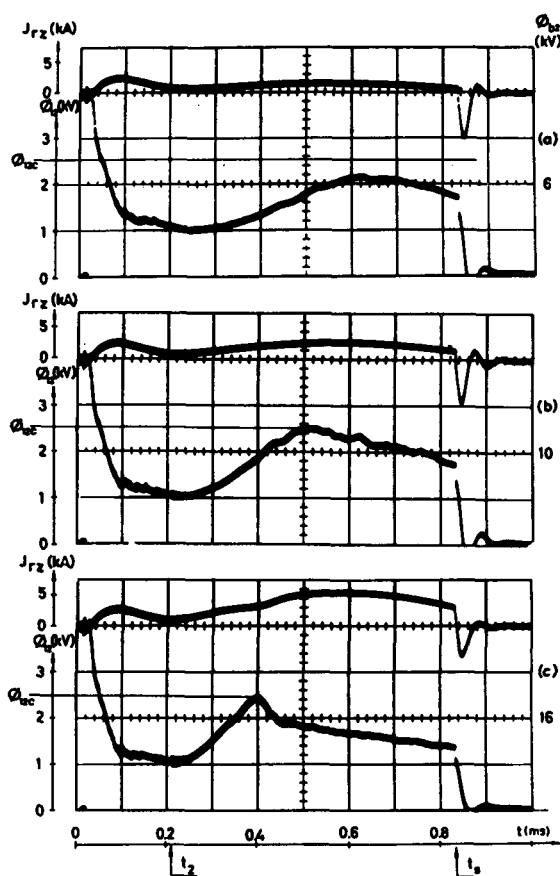


Fig. 12. Plasma current I and voltage Φ_{12} at increasing voltage Φ_{b2} of the second bank. The critical voltage Φ_{12c} is indicated by horizontal lines (Lehnert *et al.*, 1966).

level had been reached the voltage decayed in an irregular way and larger input energies only made this happen earlier. The critical voltage level, Φ_{12c} , was found to be extremely well-defined and reproducible in this experiment. Φ_{12c} was also reported to be proportional to the magnetic field. For other gases than hydrogen the level Φ_{12c} was found to be displaced (compared to the theoretically expected level) towards that for hydrogen, as in the experiment of Angerth *et al.* In the present experiment this effect was strengthened by the separation of the gases due to the strong centrifugal force component along the magnetic field lines.

The value of Φ_{12c} was reported to fit a model where the velocity distribution in the equatorial plane was parabolic centred around $\frac{1}{2}(r_{01} + r_{02})$, dropping to zero at r_{01} and r_{02} and with v_c as an upper limit along the insulator. Further, the angular velocity was constant along the magnetic field lines and

$$\phi_{12c} = \int_{r_{01}}^{r_{02}} v(r) B(r) dr.$$

Of special importance in this experiment was an investigation of the influence on the limiting voltage of the location of an auxiliary insulator – either of the ceramic tubes in Figure 11. The effect of the ceramic tubes was that by contact with the rotating plasma they introduced neutral gases at a variable distance from the axis of rotation. According to the above-mentioned model, the constraint on the velocity distribution should then be located at a predetermined position in the plasma. This was essentially verified by the experiment; the limiting voltage was observed to be inversely proportional to the mean radial distance, r_w , of the insulator. Figure 13 shows how Φ_{12c} varied with the ratio of the mean radial distance of the plasma, r_c , to the mean radial distance of the insulator, r_w , (radial ratio). r_0 was defined as the radial distance in the equatorial plane to the magnetic field line which intersects the insulator at the distance r_w from the axis. The dependence in Figure 13 was not quite linear but it was noted that Φ_{12c} increased by a factor of 2.3 when r_0/r_w increased by 2.2.

Under normal operating conditions – i.e. without the auxiliary insulators – the value of r_0/r_w was about three. At the same time the maximum velocity in the equatorial plane, as observed by means of Doppler shift measurements, was 140 km s^{-1} , implying a velocity near the critical in the vicinity of the insulator.

Summarizing the results of Lehnert *et al.* on the investigation concerning the voltage limitation in the F II-machine we can state that:

- (i) There exists a sharply defined limiting voltage – and a corresponding velocity limit – which cannot be exceeded with a moderately large power input.
- (ii) The critical voltage is approximately proportional to radial ratio r_0/r_w in the plasma.
- (iii) The results are compatible with a model for the velocity of rotation of the plasma where the velocity is zero at the plasma boundaries and limited to the critical velocity at the surface of the insulator. The plasma is rotating with constant angular velocity along the magnetic field lines, hence the velocity limit is $(r_0/r_w)v_c$ in the equatorial plane.

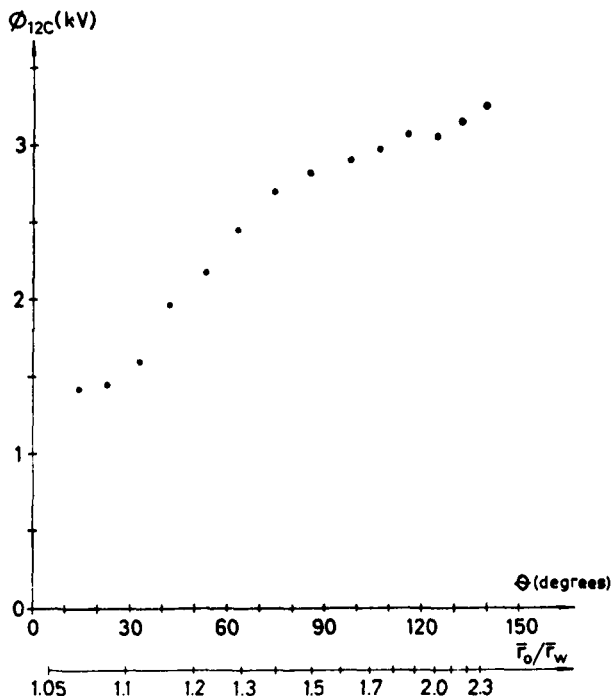


Fig. 13. Critical voltage ϕ_{12c} as a function of the angle θ of the turnable tube and the corresponding mean radial ratio $\bar{r}_o/\bar{r}_w$ (Lehnert *et al.*, 1966).

3. Plasma Guns and Shock Tubes

In some respects the plasma gun type of arrangement is advantageous compared to rotating plasmas when studying the interaction between a plasma and a neutral gas. In these experiments an ionizing front is travelling through a neutral gas and the phenomena in the front as well as in the plasma moving behind it are studied. In this situation it is much easier to observe the plasma velocity directly. Furthermore, probes, which are inserted to measure various parameters, will disturb the plasma much less since they are situated outside the plasma region until the measurement is being performed. It is also advantageous that no centrifugal effects have to be taken into account. Another important difference between this arrangement and rotating plasma experiments is that here the magnetic field lines nowhere pass from the plasma into the wall where recombination can take place and give rise to neutral gases which can interact with the plasma to limit the electric field.

The critical velocity – voltage limitation – may have been of importance (and is mentioned in the reports) in many of the experiments on plasma guns, plasma shock tubes, ionizing waves, arcs and plasma propulsion which have been performed. However, the phenomenon has only been studied in three experiments: a coaxial gun by Wilcox and Eninger (Wilcox *et al.*, 1964; Eninger, 1965), the MAST-device by Patrick and Pugh (1965) and the MAARC-device by Patrick and Schneiderman (1966).

3.1. ENINGERS COAXIAL GUN EXPERIMENT

Eninger's experimental arrangement is shown in Figure 14. It differed from a conventional coaxial gun in one essential way, viz. in the applied azimuthal magnetic field which was due to a current through the centre of the inner electrode. The applied magnetic field was always at least an order of magnitude larger than that from the

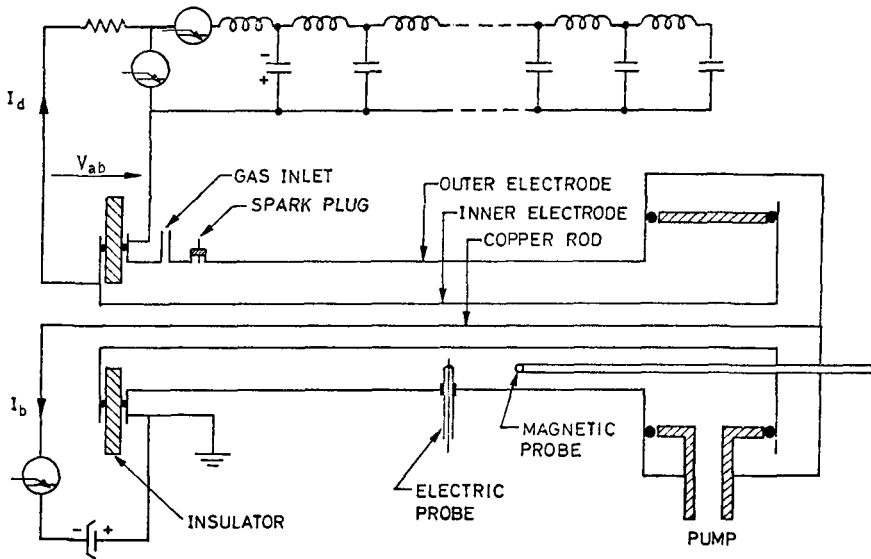


Fig. 14. Schematic drawing of plasma gun and electric circuits (Eninger, 1965).

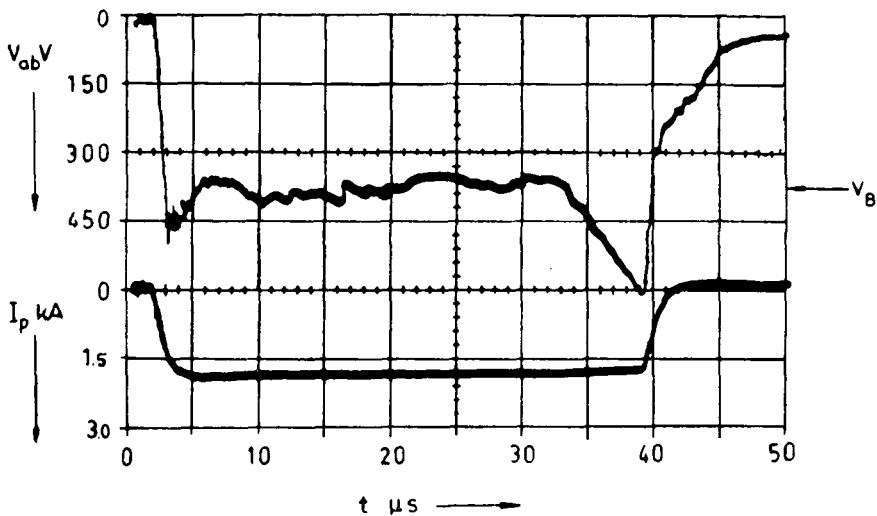


Fig. 15. Oscillograms of voltage between the electrodes (upper trace) and plasma current (lower trace) for 0.8 torr hydrogen and bias magnetic field at 1 cm radius $B_1 = 8.1$ kG. The plateau voltage V_B is indicated by an arrow (Eninger, 1965).

discharge current. Another noticeable feature was the delay-line arrangement of the condenser bank which gave a square wave current pulse of rather long duration. The typical relation between voltage and current is shown in Figure 15. As in some of the experiments described above the current was limited by a resistor R in Figure 14, of 0–1000 Ω . Other parameters of the experiment are found in Table II. The experimental results can be described in terms of four characteristic velocities, viz:

- (i) From the burning voltage of the discharge the E/B -velocity, u_b , was calculated.
- (ii) The velocity of the current sheet, u_s , was measured with magnetic probes.
- (iii) The plasma velocity, u_p , and the neutral gas velocity, u_n , were obtained from Doppler-shift measurements.

Figure 16 shows the burning voltage for different gases and magnetic fields. It is clear that the slopes of these curves are characteristic of the gas used. The radial distribution of the potential was also studied and the electric field was reported to vary as

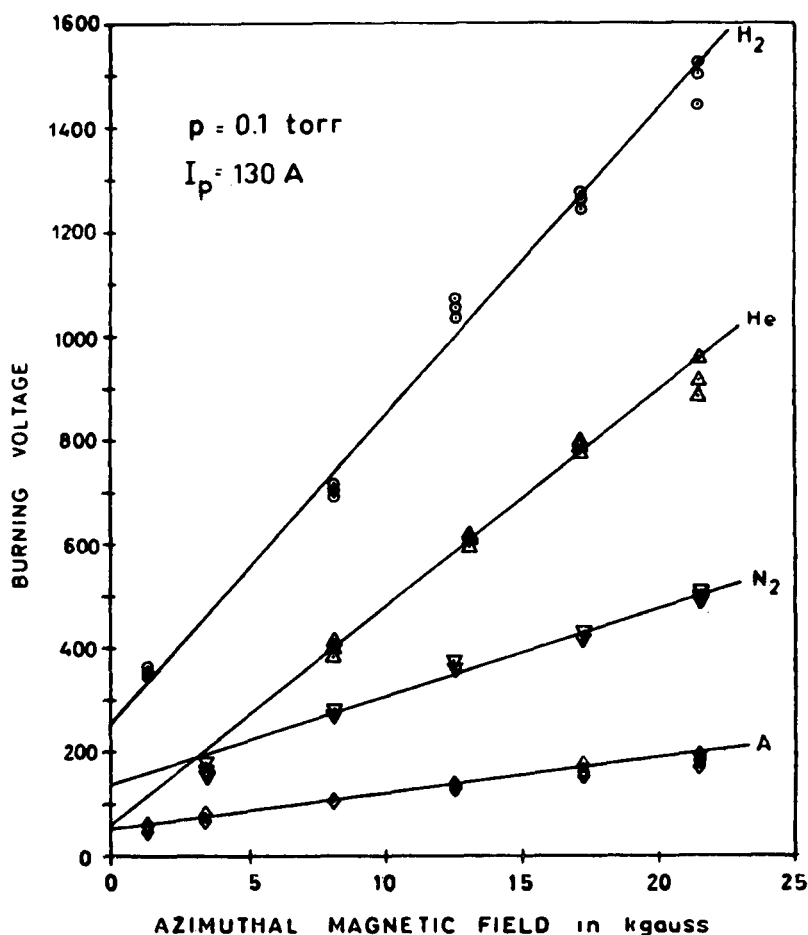


Fig. 16. Plateau voltage V_B vs bias magnetic field B_1 for H₂, He, N₂, and A (Eninger, 1965).

r^{-1} . Since B also varies as r^{-1} the ratio E/B was independent of r so that this characteristic parameter of the gas was the same everywhere in the discharge. The E/B -value was also essentially independent of the plasma current and gas pressure over four orders of magnitude of these parameters – see Figures 17 and 18. However, in the E/B vs current diagram two distinct levels were observed for H_2 and N_2 : a high one

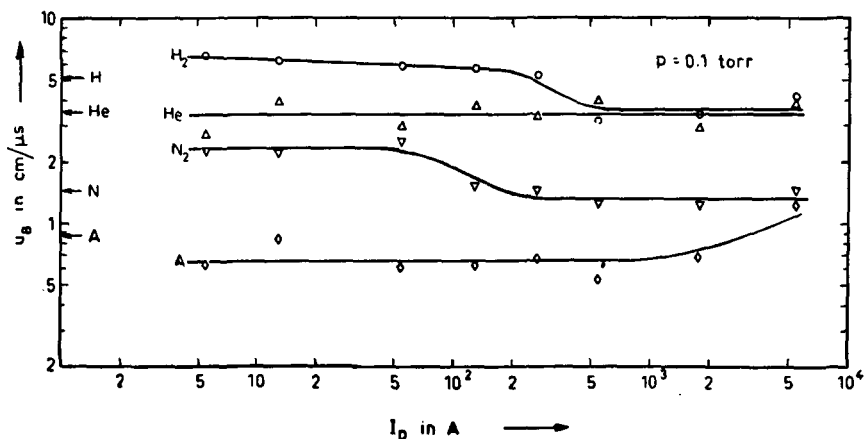


Fig. 17. E/B -velocity u_b as a function of plasma current for H_2 , He , N_2 , and A . The arrows indicate the critical velocity of the atoms (Eninger, 1965).

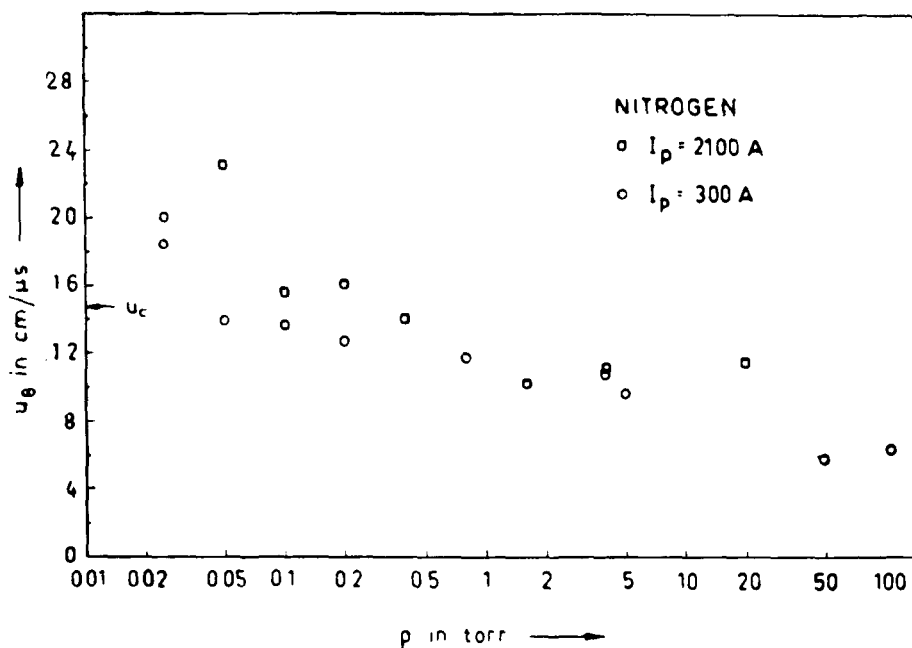


Fig. 18. Velocity u_b as a function of pressure for N_2 . The arrow indicates the critical velocity for a nitrogen atom (Eninger, 1965).

for small currents and a low one for large currents. This fact has never been explained but it was assumed to be due to the diatomic structure of these gases.

The current sheet velocity, u_s , was defined as the propagating velocity of the current layer at the front of the plasma. It was observed by measuring its time of flight between two magnetic probes. Contrary to the E/B -velocity (u_b) u_s was found to vary appreciably with pressure, current and magnetic field. It was usually lower than u_b but higher than the theoretical 'snow-plough' velocity, u_{sp} , except for the largest currents combined with the lowest pressures. The snow-plough model, which assumes that the current sheet acts as an impermeable piston ionizing and accelerating all gas in its way, yields a velocity which is

$$u_{sp} = c(\text{gas}) (IB/p)^{1/2}. \quad (4)$$

For a given geometry the constant c depends upon the gas used, and I , B and p are the plasma current, magnetic field (average) and gas pressure. The observation $u_s > u_{sp}$ implied that the current sheet was not acting as a perfect piston but as a leaky one, i.e. ions and neutrals slip through. Doppler-shift measurements of the plasma (u_p) and neutral gas (u_n) velocities were reported to show that

$$u_n \ll u_p \simeq u_s. \quad (5)$$

The main results of this experiment were discussed by means of a diagram showing u_b and u_s both plotted as a function of the u_{sp} -parameter (4) (see Figure 19). Three different regions of operation separated by the points P and Q are visualized in Figure 19. To the right of Q , for large currents and low pressures, where $u_s \simeq u_b \simeq u_{sp} > v_c$ (v_c = the critical velocity) the snow-plough model was applicable. Between P and Q there was a plateau region, where $u_s \simeq u_b \simeq v_c > u_{sp}$. Finally, to the left of P , for very high gas pressures, $u_b \simeq v_c > u_s > u_{sp}$. Thus the sheet velocity u_s is connected to the critical velocity only in the plateau region P – Q , but also to the left of P there was observed a precursor discharge travelling with the critical velocity. The constant voltage level (the E/B -velocity) agrees with the critical velocity everywhere to the left of Q .

The point P was experimentally found to be situated at a constant value of B/p , thus P moved to the left or to the right when the current was decreased or increased respectively. Furthermore, it was reported that P was situated where $(\omega_i \tau_{in} \omega_e \tau_{en})^{1/2} = 1$, ω_i and ω_e being the gyro-frequencies for ions and electrons, τ_{in} and τ_{en} the collision times for ions and electrons with neutral atoms. Thus the left-hand end of the plateau region is defined by the condition

$$\omega\tau = (\omega_i \tau_{in} \omega_e \tau_{en})^{1/2} = 1. \quad (6)$$

Eninger also noticed an interesting difference in Kerr-cell pictures of the discharge referring to the two ends of the u_b – u_s -curve. To the right of Q , i.e. for the snow-plough region, the discharge looked perfectly symmetric and smooth while to the left of P the discharge seemed highly unstable and sometimes, at low currents degenerated into a single spoke. This was interpreted as due to the resistivity connected with the ion-slip and the inapplicability of the infinite conductivity condition $\mathbf{E} + \mathbf{u}_p \times \mathbf{B} = 0$.

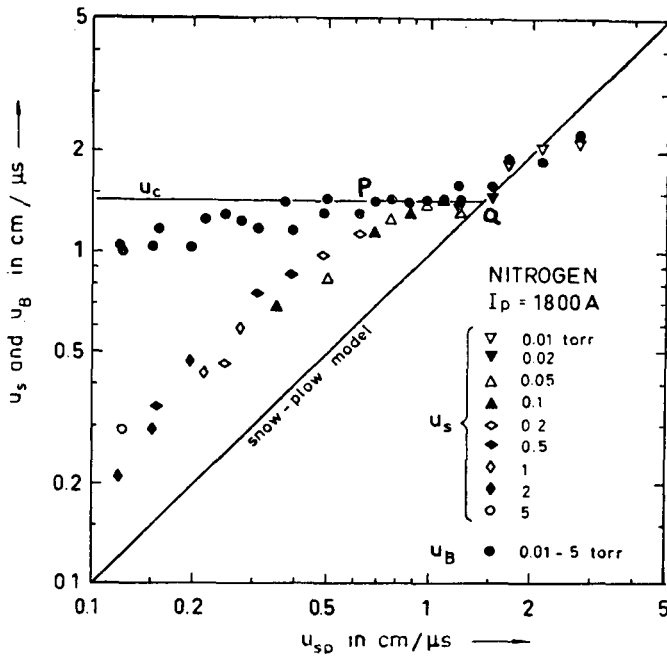


Fig. 19. Current sheet velocity u_s and E/B -velocity u_b as a function of 'snow-plow' velocity u_{sp} for nitrogen. Points P and Q indicate transitions between different regimes of operation (Eninger, 1965).

Recent experiments by Axnäs (1972) have shed new light on this experiment. As in some of the rotating experiments, he applied a step increase of the discharge current. The conditions before and after the current increase were measured. In the plateau region the main effect was a very fast increase of the plasma density while the velocity remained essentially unchanged. The new equilibrium was reached in less than $1 \mu\text{s}$. A lower limit to the ionization frequency is obtained by measuring the relaxation time of a burning voltage transient, caused by the sudden increase of the discharge current. The experimental results are compared with theoretical predictions of electron heating mechanisms, which are classical in the sense that only binary collision processes are considered. It is found that the experimentally measured ionization frequency is larger by 1–2 orders of magnitude than that theoretically predicted for classical processes. Collective processes involving fast heating of the electrons could explain the phenomena (cf. the experiment by Danielsson below).

3.2. THE MAST-EXPERIMENT

The Magnetic Annular Shock Tube of the Avco-Everett Research Laboratory was essentially similar to the plasma gun used by Wilcox and Eninger. In one version of it, Keck (1962), there was no magnetic field in the plane of the ionizing front and in this version nothing related to the voltage limitation or critical velocity was observed. In another version, Patrick (1959), an azimuthal magnetic field was added in the same

way as in Eninger's experiment. Patrick and Pugh (1965) reported from this experiment that the velocity of the ionizing front as well as the value of E/B was limited to the order of v_c . In this experiment also the ratio E/B was independent of position in the discharge.

3.3. THE MAARC-EXPERIMENT

The Magnetic Annular ARC of the Avco-Everett Research Laboratory is actually a device to study steady state propulsion problems. The main parameters are found in Table II. For all four propellants studied, a burning voltage dependence which could be expressed by the formula $V = V_0 + v_c B d$ was observed. In the formula, V_0 is the total electrode sheath voltage, v_c the critical velocity, B the magnetic field and d the length of the discharge normal to the magnetic field. In this experiment, contrary to the rest of the experiments discussed, the applied magnetic field was essentially axial and parallel to the plasma stream. The magnetic field was, however, not purely axial, there being a varying degree of divergence of the field. Not unexpectedly, the burning voltage was also reported to depend on the geometry of the magnetic field in addition to its magnitude. It is possible that the voltage is connected to a rotation of the plasma around the axis of the coaxial system. There could then be a relative velocity across the magnetic field which is connected to the critical velocity and voltage limitation as in the other experiments of the present review. There is however no experimental evidence as to such a rotation here, but this has been observed in other arc experiments (Kogelschatz, 1969; Kribel *et al.*, 1969) and suggested in a few more (see e.g. Blackstock *et al.*, 1968).

The axial velocity of the plasma stream was not measured directly in the MAARC-experiment; by means of thrust measurements it was estimated to be 30 km s^{-1} for hydrogen or about 60% of v_c . The exact relation of this velocity to the critical velocity in the gun is unclear.

The effect of the critical velocity on MHD-arcs is mentioned in a great number of reports. For a number of reasons these will not be discussed here:

(i) As far as the critical velocity is concerned the findings appear questionable or at least very unclear.

(ii) The effect is generally not studied systematically.

(iii) In terms of Figure 19 the arc-experiments are located to the left of the point P ($\omega\tau < 1$), where no connection between the critical velocity and the plasma velocity is expected.

(iv) The arc-experiments are thoroughly discussed elsewhere, see Devillers (1971), or Devillers and Burton (1970).

4. Direct Collision Between A Plasma and a Neutral Gas

In a recent experiment (Danielsson, 1970) a completely new approach to the interaction between a plasma and a neutral gas was taken. Several of the relevant experimental characteristics were the reverse of those of the experiments described above. The

plasma was fully ionized and had a super-critical velocity prior to the commencement of the interaction. There were no external electric connections to the interaction region and hence the macroscopic net current was zero.

The experimental arrangement is shown in Figure 20. The hydrogen plasma was generated and accelerated in an electrodeless plasma gun (a conical theta pinch) and flowed into a drift tube along a magnetic field. The direction of the magnetic field

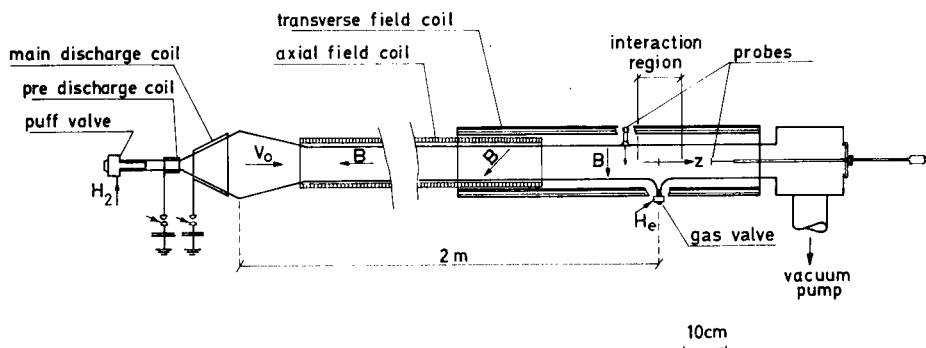


Fig. 20. Experimental arrangement of Danielsson (1969).

changed gradually from axial to transverse while the magnitude remained unchanged ($0.1-0.5 \text{ Vs m}^{-2}$). As the plasma flowed along the drift tube much of it was lost but a polarization electric field was developed and a plasma of about $3 \times 10^{17} \text{ m}^{-3}$ drifted across the transverse magnetic field with a typical velocity of 300 km s^{-1} . In the region of the transverse magnetic field the plasma penetrated a small cloud of neutral gas, usually helium, released from an electromagnetic valve. This helium cloud had an axial depth of 5 cm and a density of 10^{20} m^{-3} ($3 \mu\text{Hg}$) at the time of arrival of the plasma. The remainder of the system was under high vacuum ($3 \times 10^{-7} \text{ mm Hg}$). Under these conditions the mean free paths for all important, direct, binary collisions were much more than 5 cm so that the interaction according to common terminology was classified as collisionless.

Double probes were used to measure the polarization electric field $\mathbf{E}$. When the upstream plasma velocity v_0 was high enough (see below) it was found that $\mathbf{E}$ decreased from its upstream value to a relatively low downstream value over distances of one or two centimetres. Although the connection between $\mathbf{E}$ and the plasma velocity $\mathbf{v}$ is not straightforward in the centre of the interaction region, where the scale length for variations in $\mathbf{E}$ and the proton Larmor radius are of the same order, the upstream and downstream $\mathbf{E}$ is relatively uniform so that the plasma velocity can be deduced from $\mathbf{v} = \mathbf{E} \times \mathbf{B} / B^2$. In other words the interaction with the neutral gas cloud strongly decelerated the plasma beam over distances of a few centimetres (see Figure 21).

The relationship between the upstream velocity v_0 and the downstream velocity 1 cm behind the centre of the gas cloud, $v_{z=1}$, is shown in Figure 22. It was found

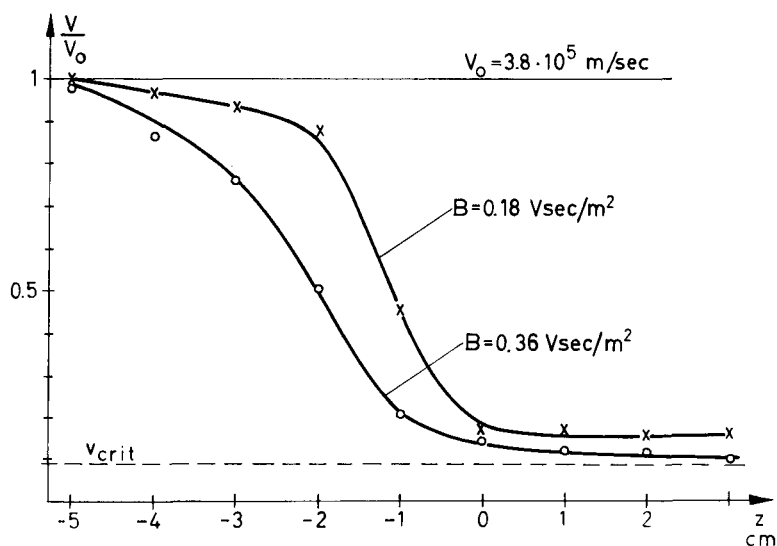


Fig. 21. Plasma velocity at different points in the neutral gas cloud. $z = 0$ is the center of the cloud (Danielsson, 1969).

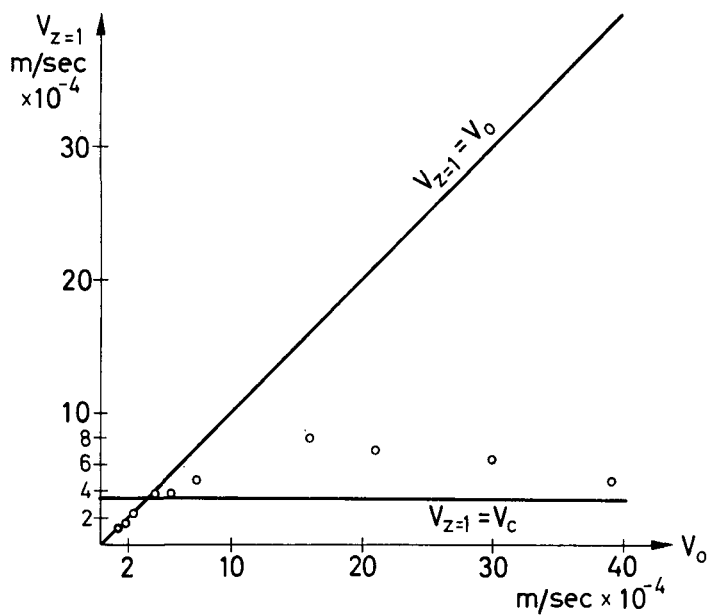


Fig. 22. Plasma velocity in the neutral gas cloud as function of the plasma vacuum velocity (Danielsson, 1969).

that for very low v_0 the plasma was not retarded as it penetrated the gas cloud, whereas for $v_0 > 40 \text{ km s}^{-1}$ the relative retardation increased with v_0 . The critical velocity for helium, $v_c = 35 \text{ km s}^{-1}$, is also shown in Figure 22. The plasma velocity was never reduced below v_c . Measurements with capacitatively coupled probes placed outside the drift tube indicated that substantial numbers of helium ions were formed in the interaction. Absolute emission intensity measurements and 4 mm microwave measurements also indicated a substantial production of helium ions. It is worth noting here that if the electron motion were adiabatic then the constant value of n/B together with the constant value of B implies that any increase in the electron density n is due to ionization.

Before the interaction no radiation was detected from either the gas or the plasma (fully ionized protons) and measurements in the plasma gun region indicated an electron temperature $\sim 5 \text{ eV}$. However, when the plasma penetrated the gas cloud, emission was observed with a fast rise time ($< 0.5 \mu\text{s}$), even from energetic levels of He^+ . The low electron density prevailing here then implied that stepwise processes could not be responsible for the rapid onset of the emission, and so the line intensities were interpreted in terms of the cross-section for single step electron excitation of the particular line. Measurements of the relative intensity at $4686 \text{ \AA}$ in He^+ and $4713 \text{ \AA}$ in He thus indicated a group of electrons with a characteristic energy of about 85 eV (assuming a monoenergetic group to be responsible for the emission). Alternatively a Maxwellian distribution with a temperature corresponding to 70 eV would lead to the same observed intensity ratios. These intensity measurements were made with $v_0 = 400 \text{ km s}^{-1}$ and a corresponding proton energy of 840 eV . It should also be noted that several tests were made to check that this large and rapid increase in electron energy was in fact caused by the interaction and not by some other extraneous effect.

It was found that, provided the gas density was above roughly $5 \times 10^{19} \text{ m}^{-3}$, gas density variations had little effect on the plasma retardation. On the other hand the retardation quickly disappeared when the density was reduced below $2 \times 10^{19} \text{ m}^{-3}$. Similarly it was found that although a magnetic field was necessary for the retardation to occur, doubling its strength from 0.18 to 0.36 Vs m^{-2} had relatively little effect on the retardation.

Consideration of the typical gas density and *undisturbed* plasma parameters led to the conclusion that simple binary collisions between the undisturbed plasma and gas could not account for the observed retardation. On the other hand, estimates of the production rate of helium ions produced by the observed *high energy* electrons were large enough to account for the retardation. A theoretical description of the interaction must then be along the lines of a collective interaction between the plasma and the gas which leads to the production of high energy electrons. We should note here how well the retardation observed in this experiment supports Alfvén's critical velocity hypothesis (Alfvén, 1954).

So far Danielsson's experiment has demonstrated that even in a situation where the primary collisions are negligibly few there may be a very strong interaction

between a moving plasma and a stationary gas. This interaction is active above 40 km s^{-1} in helium ($v_c = 35 \text{ km s}^{-1}$) and it leads to

- (i) local heating of the electrons;
- (ii) ionization of the neutral gas;
- (iii) retardation of the plasma stream.

5. Summary and Discussion of the Experiments

5.1. EXPERIMENTAL OBSERVATIONS

The present situation is that a gas-dependent voltage limitation of an electric discharge across a magnetic field in a partially ionized plasma has been reported from a considerable number of experiments. In two cases (Danielsson, 1970, and Axnäs, 1972) this has been directly connected to an enhanced interaction between the plasma and the neutral gas. In others this connection is indicated by the fact that the voltage limitation is independent on the power input. In a number of situations the voltage limitation is also experimentally connected to velocity limitation. As long as the plasma is only partially ionized and the neutral component remains at rest it is also independent of discharge current and plasma density. The limiting voltage increases linearly with the magnetic field according to (1) as

$$V_b = V_0 + v_c dB.$$

The constant v_c in (1), which has the dimension of a velocity, is characteristic of the neutral gas present. (Mixtures of gases have been omitted from this review.) It has a value which very closely agrees with the relation (3) namely,

$$\frac{1}{2} M v_c^2 = e V_i.$$

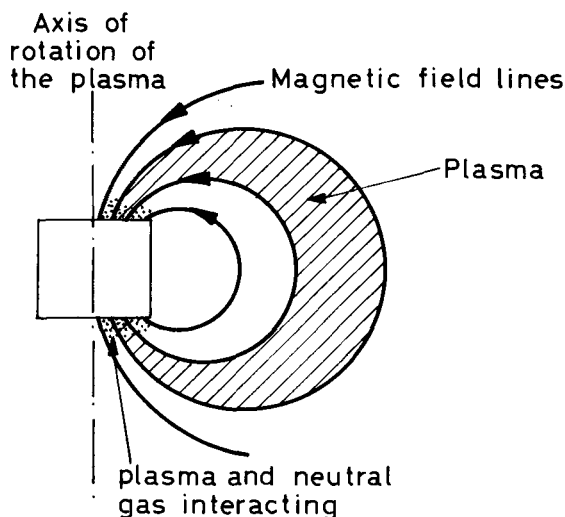


Fig. 23. The velocity of the entire plasma is limited by the interaction with neutral gas near the wall when magnetic field lines intersect the wall.

The condition of partial ionization has in the past caused some confusion; in some but not all experiments a fully ionized plasma has also been subject to the voltage limitation. The decisive condition seems to be whether or not the plasma is confined by walls intersected by the magnetic field lines (compare the difference between the rotating plasma and coaxial gun experiments). Due to the isorotation effect a thin boundary layer of neutral gas on the walls can interact with a rotating plasma to limit the velocity of the entire plasma volume when magnetic field lines connect the boundary layer to the plasma (see Figure 23). On the other hand, in the coaxial gun experiments, no magnetic field lines tie the plasma to the walls and the critical velocity can be exceeded at complete ionization.

5.2. VOLTAGE LIMITATION – CRITICAL VELOCITY

In most of the discussed experiments the primary observation refers to the voltage over the plasma across the magnetic field. Essentially the dependence (1) or (2) is observed in all experiments where a limiting voltage is reached. Due to the fact that v_c has the dimension of a velocity, which is connected to the experimentally observed limiting voltage according to (1), the concept of a critical velocity is equivalent to that of a voltage (– or rather – E/B) limitation. The velocity then has the advantage over the voltage that it is independent of one more parameter, viz. the magnetic field. Thus, basically what is implied by the concept ‘critical velocity’ is in reality a critical value of E/B . In some of the experiments, where the ion Hall parameter is large, $(\omega\tau)_i \gtrsim 10$, the velocity of the plasma is found to agree very well with the value of E/B . This is true at the limiting voltage (1) where $v \approx v_c$, as well as at both higher and lower velocities. This establishes the connection between the observed voltage and a real velocity of the plasma. However, in situations where the ion Hall parameter is small, there is no direct correlation between the plasma velocity and the E/B -value.

5.3. AVOIDING THE CRITICAL VELOCITY LIMITATION

As already mentioned, the critical velocity appears in most cases as an undesired limitation of the energy which can be transferred to a plasma. Many attempts have been made to avoid this limitation; some of these have been fairly successful, others have failed completely. As shown by Lehnert *et al.* (1966) the crucial point is whether the plasma is confined by any material walls intersected by the magnetic field lines. If this is the case it does not seem possible to avoid the limitation; the wall seems always to be able to eject a sufficient amount of gas to limit the kinetic energy of the plasma.

One way to increase the velocity limit without removing this condition is demonstrated by Lehnert *et al.* (1966), viz. to increase the radial ratio of a rotating plasma (see Section 2.3).

Acknowledgements

The author is indebted to Drs John Sherman and Lars Block for frequent discussions on the phenomena under discussion.

References

- Alfvén, H.: 1954, *On the Origin of the Solar System*, Ch. III, Oxford University Press.
- Alfvén, H.: 1960, *Rev. Mod. Phys.* **32**, 710.
- Angerth, B., Block, L., Fahleson, U., and Soop, K.: 1962, *Nucl. Fusion Suppl.* Pt 1, 39.
- Axnäs, A.: 1972, Department of Electron and Plasma Physics, Royal Institute of Technology, Research Report 72-31.
- Baker, D. A. and Hammel, J. E.: 1961, *Phys. Fluids* **4**, 1549.
- Baker, D. A., Hammel, J. E., and Ribe, F. L.: 1961, *Phys. Fluids* **4**, 1534.
- Bannenberg, J. G., Brakenhoff, G. J., and Kistemaker, J.: 1966, *1st Europ. Conf. Controlled Fusion and Plasma Phys.*, Munich, and F.O.M. Inst., Amsterdam, Rept. 22460.
- Bannenberg, J. G., Insinger, F. G., Rasmussen, C. E., and Kistemaker, J.: 1963, *Proc. 6th Int. Conf. Ioniz. Phenom. Gases* (Paris) **2**, 393.
- Barber, P. B., Pilcher, M. L., Swift, D. A., and Tozer, B. A.: 1963, *Proc. 6th Int. Conf. Ioniz. Phenom. Gases* (Paris) **2**, 395.
- Bergström, J., Holmberg, S., and Lehnert, B.: 1962, *Nucl. Fusion Suppl.* Pt. 2, 705.
- Bergström, J., Holmberg, S., and Lehnert, B.: 1963, *Arkiv Fysik* **25**, 49.
- Bergström, J., Holmberg, S., and Lehnert, B.: 1966, *Plasma Physics and Controlled Nuclear Fusion Research* **1**, 341, I.A.E.A., Vienna.
- Blackstock, A. W., Fradkin, D. B., Roehling, D. J., and Stratton, T. F.: 1968, *J. Appl. Phys.* **39**, 3201.
- Boyer, K., Hammel, J. E., Longmire, C. L., Nagel, D., Ribe, F. L., and Riesenfeld, W. B.: 1958, *Proc. 2nd Int. Conf. Pieceful Uses Atomic Energy* **31**, 319. U.N., Geneva.
- Bratenahl, A., Kunkel, W. B., Baker, W. R., etc.: 1959, University of California Lawrence Radiation Lab., U.C.R.L. Report 9002.
- Bratenahl, A., Kunkel, W. B., Baker, W. R., etc.: 1960, University of California Lawrence Radiation Lab., U.C.R.L. Report 9106 and 9243.
- Danielsson, L.: 1970, *Phys. Fluids* **13**, 2288.
- Devillers, P. J.: 1971, Dissertation at the department of Engineering Sciences, University of California at San Diego.
- Devillers, P. J. and Burton, R. L.: 1970, AIAA-paper 70-1096 (Bull. AIAA).
- Eninger, J.: 1965, *Proc. 7th Int. Conf. Ioniz. Phenom. Gases* (Belgrade) **1**, 520.
- Fahleson, U. V.: 1961, *Phys. Fluids* **4**, 123.
- Halbach, K., Baker, W. R., and Layman, W. R.: 1962, *Phys. Fluids* **5**, 1482.
- Keck, J. C.: 1962, *Phys. Fluids* **5**, 630.
- Kogelschatz, U.: 1959, AIAA-Paper 69-110.
- Kribel, R., Eckdahl, C., and Lovberg, R.: 1969, A.I.A.A.-Paper 69-245.
- Lehnert, B.: 1966, *Phys. Fluids* **9**, 774.
- Lehnert, B., Bergström, J., and Holmberg, S.: 1966, *Nucl. Fusion* **6**, 231.
- Patrick, R. M.: 1959, *Phys. Fluids* **2**, 589.
- Patrick, R. M. and Pugh, E. R.: 1965, *Phys. Fluids* **8**, 636.
- Patrick, R. M. and Schneiderman, A. M.: 1966, *AIAA J.* **4**, 283.
- Shvets, O. M., Ovchinnikov, S. S., and Tarasenko, V. F.: 1966, in K. D. Sinel'nikov (ed.), *Plasma Physics and Controlled Thermonuclear Fusion*, Jerusalem, p. 388.
- Shvets, O. M., Ovchinnikov, S. S., Tarasenko, V. F., and Tolok, V. T.: 1963, *Fizika Plazmy Upravlyae-mogo Termoyadernogo Sinteza* **3**, 184, Izdatel'stvo A. N. Ukr. S.S.R., Kiev.
- Simon, A.: 1959, G. E. Research Lab. Rept. 59-RL-2322 E.
- Wilcox, J. M., Pugh, E., Dattner, A., and Eninger, J.: 1964, *Phys. Fluids Suppl.* **7**, S51.